

# Regression Diagnostics

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# Outline

- 1 Regression Model and Its Assumptions
- 2 Model Diagnostics
- 3 Residual Analysis
- 4 Regression Performance Measures

# Regression Model and Its Assumptions

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# The multiple linear regression model

- We observe  $n$  pairs  $(\mathbf{x}^{(i)}, y^{(i)})$ ,  $i = 1, \dots, n$ , with  $p$  features in each  $\mathbf{x}^{(i)}$
- The model, written for each observation:

$$y^{(i)} = \beta_0 + \beta_1 x_1^{(i)} + \dots + \beta_p x_p^{(i)} + \varepsilon^{(i)}$$

- In matrix form,  $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ , where  $\tilde{\mathbf{X}}$  is the  $n \times (p + 1)$  design matrix whose  $i$ -th row is  $(1, x_1^{(i)}, \dots, x_p^{(i)})$
- The coefficients are estimated by least squares; inference (standard errors, intervals, tests) needs the assumptions on the next two slides

# Assumptions of the MLR model (1/2): the mean and the features

- **Linearity:** the mean of  $y$  is linear in the coefficients,  
$$\mathbb{E}[y^{(i)} \mid \mathbf{x}^{(i)}] = \beta_0 + \sum_{j=1}^p \beta_j x_j^{(i)}$$
- **Zero-mean errors:**  $\mathbb{E}[\varepsilon^{(i)} \mid \tilde{\mathbf{X}}] = 0$ , so the errors are unrelated to the features
- **No perfect collinearity:**  $\tilde{\mathbf{X}}$  has full column rank  $p + 1$ , so  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$  exists and the coefficients are uniquely determined

## Assumptions of the MLR model (2/2): the errors

- **Uncorrelated errors:**  $\text{Cov}(\varepsilon^{(i)}, \varepsilon^{(k)}) = 0$  for  $i \neq k$
- **Constant variance (homoscedasticity):**  $\text{Var}(\varepsilon^{(i)}) = \sigma^2$  for every  $i$
- **Normal errors:**  $\varepsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$ ; needed for exact small-sample intervals and tests, not for the estimates themselves

## These assumptions may not hold

- Real data often violate one or more of them: the true relation is curved, errors are correlated (over time or within units), the spread of the errors changes with the level of  $y$ , a few observations are unusual, or features are nearly collinear
- Consequences: wrong predictions (non-linearity); wrong standard errors, intervals and  $p$ -values (correlated errors, non-constant variance, non-normal errors); unstable coefficients (collinearity); a few points distorting the whole fit (outliers, high leverage)
- So after fitting, the assumptions have to be checked, not assumed

# Model Diagnostics

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# What is model diagnostics?

- **Model diagnostics:** checks made after fitting to see whether the model assumptions hold, and whether the fit depends on a few observations or on redundant features
- The checks use quantities computed from the fit: residuals, studentized residuals, leverages and variance inflation factors
- Three questions: Is the linear model adequate? Which observations are unusual? Are some features redundant?
- Diagnostics examine whether the fit can be trusted; performance measures (later in this lecture) examine how well it fits

# Coefficient estimates and fitted values

- Least squares chooses  $\beta$  to minimize

$$\text{RSS} = \sum_{i=1}^n (y^{(i)} - \beta_0 - \sum_{j=1}^p \beta_j x_j^{(i)})^2, \text{ which gives}$$

$$\hat{\beta} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

- The fitted values are obtained by plugging  $\hat{\beta}$  into the model:

$$\hat{\mathbf{y}} = \tilde{\mathbf{X}}\hat{\beta}, \quad \hat{y}^{(i)} = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_j^{(i)}$$

# The hat matrix

- Substituting  $\hat{\beta}$  into  $\hat{y} = \tilde{\mathbf{X}}\hat{\beta}$  expresses the fitted values as a linear map of the labels:

$$\hat{y} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top$$

- $\mathbf{H}$  is the **hat matrix**:  $n \times n$ , symmetric and idempotent ( $\mathbf{H}^2 = \mathbf{H}$ ), with  $\text{tr}(\mathbf{H}) = p + 1$
- $\mathbf{H}$  depends only on the features in  $\tilde{\mathbf{X}}$ , never on  $\mathbf{y}$

# The residual vector

- The residual for observation  $i$  is the gap between the observed and fitted value,  $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$ ; in vector form  $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$
- Using  $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ , the residual vector is a linear function of  $\mathbf{y}$ :

$$\mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

- The residuals estimate the unobservable errors:  $\varepsilon^{(i)}$  is never known,  $e^{(i)}$  is computed from the data

# The residual vector is a random vector

- $\mathbf{e}$  is a function of the random errors  $\varepsilon$ , so each  $e^{(i)}$  is a random variable and  $\mathbf{e}$  is a random vector
- Its mean is  $\mathbb{E}[\mathbf{e}] = \mathbf{0}$ ; its spread is described by the **variance-covariance matrix**

$$\text{Cov}(\mathbf{e}) = \mathbb{E}[(\mathbf{e} - \mathbb{E}[\mathbf{e}])(\mathbf{e} - \mathbb{E}[\mathbf{e}])^\top] = \sigma^2(\mathbf{I} - \mathbf{H})$$

- $\text{Cov}(\mathbf{e})$  is  $n \times n$  and symmetric
- Diagonal entries are the individual variances:  $\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$
- Off-diagonal entries are the pairwise covariances:  
 $\text{Cov}(e^{(i)}, e^{(k)}) = -\sigma^2 H_{ik}$  for  $i \neq k$
- The errors are uncorrelated, but the residuals generally are not, because  $H_{ik} \neq 0$

# The variance-covariance matrix of the residuals

- $\text{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})$  is an  $n \times n$  symmetric matrix

$$\text{Cov}(\mathbf{e}) = \sigma^2 \begin{pmatrix} 1 - h^{(1)} & -H_{12} & -H_{13} & \cdots & -H_{1n} \\ -H_{21} & 1 - h^{(2)} & -H_{23} & \cdots & -H_{2n} \\ -H_{31} & -H_{32} & 1 - h^{(3)} & \cdots & -H_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -H_{n1} & -H_{n2} & -H_{n3} & \cdots & 1 - h^{(n)} \end{pmatrix}$$

- Blue diagonal: the individual variances  $\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$
- Black off-diagonal: the pairwise covariances  $\text{Cov}(e^{(i)}, e^{(k)}) = -\sigma^2 H_{ik}$  for  $i \neq k$

## Where the hat matrix is used: leverage

- The diagonal of  $\text{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})$  (previous slide) gives the variance of each residual:

$$\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)}), \quad h^{(i)} = H_{ii}$$

- $h^{(i)}$  is the **leverage** of observation  $i$ ; it depends only on the features, and the leverages add up to  $\sum_{i=1}^n h^{(i)} = p + 1$
- Even with constant error variance, residuals have unequal variances: high-leverage observations have smaller residual variance, because the fit is pulled towards them

## Where the hat matrix is used: studentized residuals

- Raw residuals have unequal variances and carry the units of  $y_i$ ; dividing by the estimated standard deviation fixes both:

$$r^{(i)} = \frac{e^{(i)}}{\hat{\sigma}\sqrt{1 - h^{(i)}}}, \quad \hat{\sigma} = \text{RSE}$$

- The leverage  $h^{(i)} = H_{ii}$  comes from the hat matrix, so the studentized residual uses both the residual and the hat matrix
- Each  $r^{(i)}$  has variance close to 1 and no units, so residuals can be compared across observations and across models
- With normal errors about 99.7% of the  $r^{(i)}$  fall within  $\pm 3$ : an observation with  $|r^{(i)}| > 3$  is unusual, although among 400 observations one such value is expected by chance
- The studentized residual is also called the standardized residual; in R it is computed by `rstandard()`



# Residual Analysis

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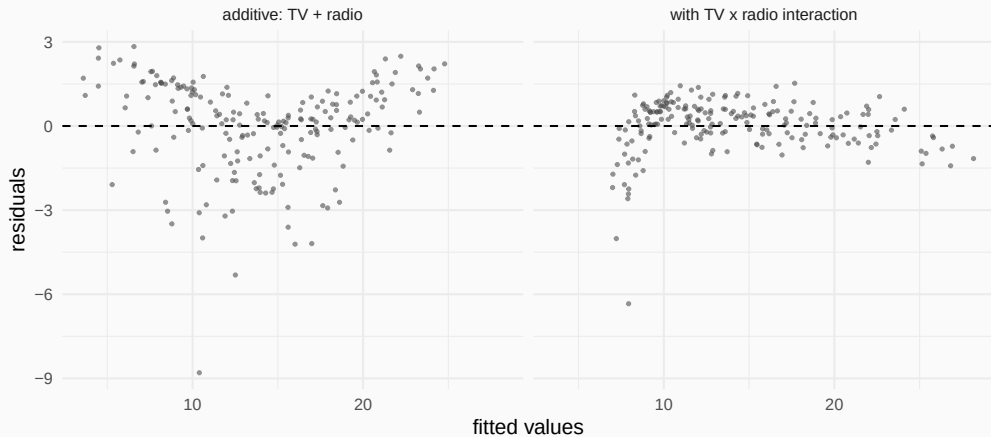
# What is residual analysis?

- **Residual analysis:** examining the residuals  $e$  (or transformed versions of them) to judge whether the assumptions of the model hold
- If the assumptions hold, the residuals should look like random noise: centred at zero, with constant spread, no pattern against the fitted values, the features, or the observation order
- Main plots: residuals against fitted values, residuals against each feature, residuals against observation order, and the normal QQ plot of the residuals
- Observation-level quantities (leverage and studentized residuals) refine the check; they need the hat matrix, shown next

## Problem 1: non-linearity, what to check

- **Plot:** residuals  $e^{(i)}$  against fitted values  $\hat{y}^{(i)}$ , with a horizontal line at zero
- **If** the points follow a curve (U-shape or inverted U), **then** the linearity assumption is violated
- **If** the points scatter randomly around zero with no pattern, **then** linearity is plausible
- **Consequence:** predictions are biased in the ranges where the curve lies, and inference is unreliable
- **Remedies:** transform features ( $\log x$ ,  $\sqrt{x}$ ), add interaction terms, or fit a more flexible model

# Problem 1: non-linearity, example



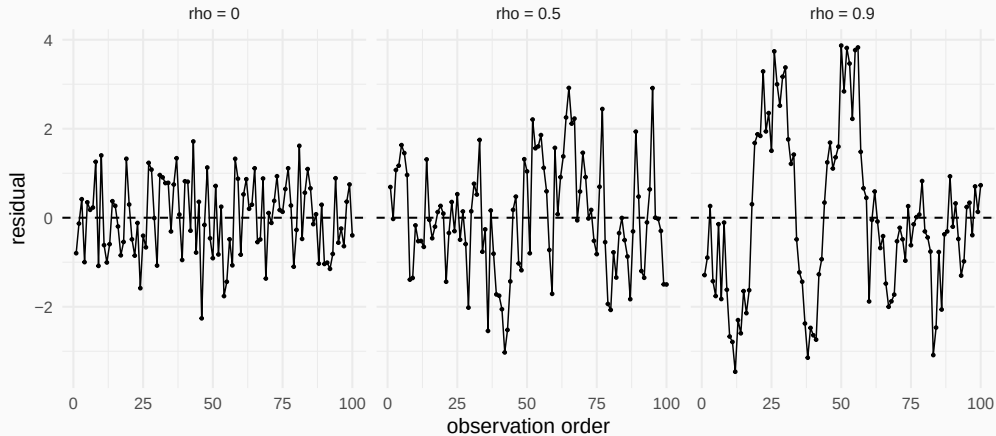
## Problem 1: non-linearity, example (cont.)

- Dataset: Advertising,  $n = 200$  markets (sales in thousands of units; TV and radio budgets in thousands of dollars)
- Left panel: additive model,  $\text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{radio} + \varepsilon$ ; right panel: the same model with a  $\text{TV} \times \text{radio}$  term
- Additive model: residuals are positive at the smallest and largest fitted values and negative in between, a U-shape that signals missing synergy between TV and radio
- With the interaction the curvature weakens and RSE falls from 1.68 to 0.94; a few large negative residuals remain

## Problem 2: correlated errors, what to check

- **Plot:** residuals  $e^{(t)}$  against observation order  $t$ , with a horizontal line at zero
- **If** neighbouring residuals are alike, with long runs of the same sign, **then** the uncorrelated-errors assumption is violated (successive errors are positively correlated)
- **If** the signs change at random from one observation to the next, **then** there is no evidence of correlation
- **Consequence:** standard errors are too small, so intervals are too narrow and  $p$ -values too small
- **Remedy:** time-series or mixed models, which are outside the scope of this lecture

## Problem 2: correlated errors, example



## Problem 2: correlated errors, example (cont.)

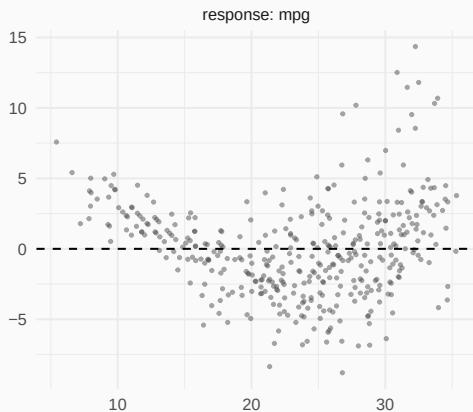
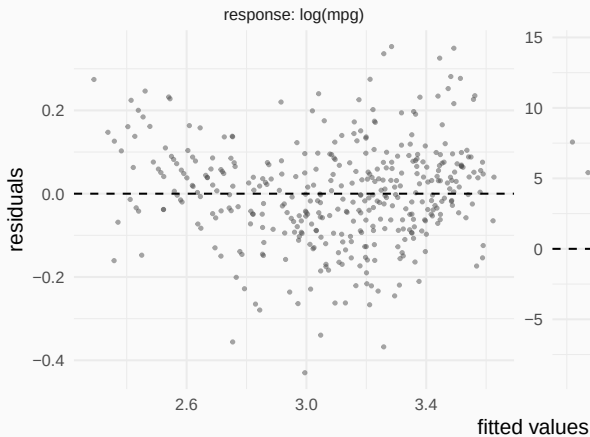
- The above plots are from simulated datasets
- In the left plot, there is no autocorrelation among the errors, i.e., the observations are independent
- In the middle plot, autocorrelation is moderate (0.5). The residuals get stuck on one side of the horizontal axis
- In the right plot, autocorrelation is very high (0.9), the residuals get stuck on one side of the horizontal for much longer



## Problem 3: non-constant variance, what to check

- **Plot:** residuals  $e^{(i)}$  against fitted values  $\hat{y}^{(i)}$
- **If** the vertical spread grows or shrinks with the fitted value (a funnel), **then** the constant-variance assumption is violated
- **If** the vertical spread is about the same across the whole range, **then** constant variance is plausible
- **Consequence:** standard errors, intervals and tests become unreliable
- **Remedy 1:** transform  $y$  ( $\log y$  or  $\sqrt{y}$ )
- **Remedy 2:** weighted least squares

# Problem 3: non-constant variance, example



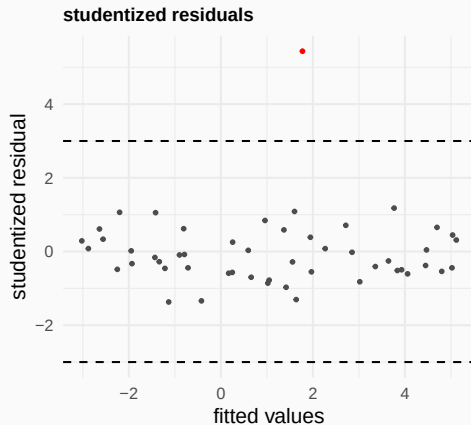
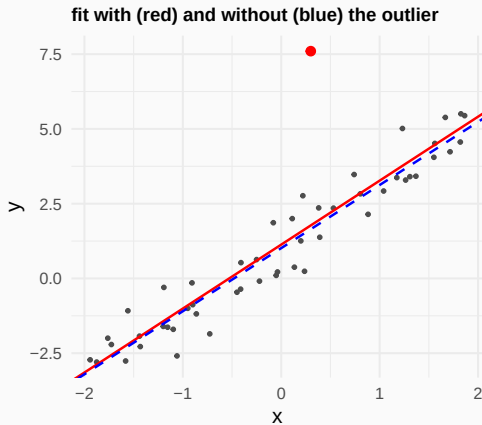
## Problem 3: non-constant variance, example (cont.)

- Dataset and model: 'Auto' dataset,  $n = 392$  cars;  
 $\text{mpg} = \beta_0 + \beta_1 \text{horsepower} + \beta_2 \text{weight} + \beta_3 \text{year} + \varepsilon$
- Left panel:  $\log(\text{mpg})$  as response; right panel: mpg as response; the y-scales are free
- The mpg residuals fan out as the fitted value grows; their standard deviation grows from 2.35 to 2.48, 3.47 and 3.88 across the four quartiles of the fitted values
- For  $\log(\text{mpg})$  the spread is nearly constant (0.116, 0.117, 0.130, 0.107), so the coefficients describe multiplicative effects

## Problem 4: outliers, what to check

- **Plot:** studentized residuals  $r^{(i)}$  against fitted values  $\hat{y}^{(i)}$ , with horizontal lines at  $\pm 3$
- **If** a point lies beyond  $\pm 3$ , **then** it is a suspected outlier: its error is far outside the pattern shown by the rest of the data
- **Consequence:** an outlier inflates RSE and lowers  $R^2$ , and it can pull the fit towards itself
- **Action:** investigate the point (data-entry error or a genuine case) before deleting anything; refit and compare

## Problem 4: outliers, example



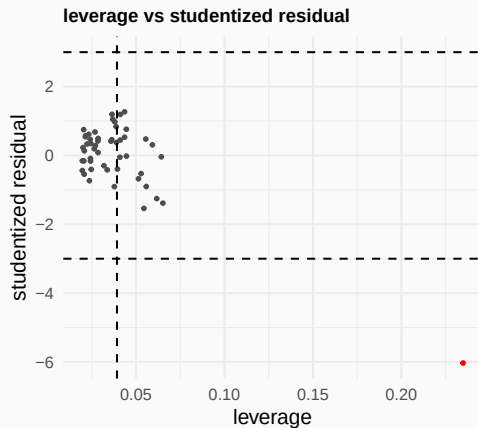
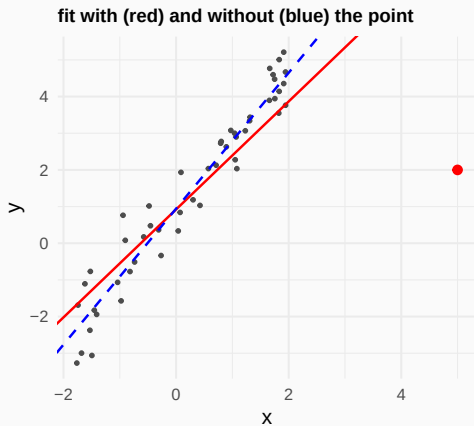
## Problem 4: outliers, example (cont.)

- Dataset (simulated): 50 points with  $x \sim U(-2, 2)$  and  $y = 1 + 2x + \varepsilon$ ,  $\varepsilon \sim N(0, 0.7^2)$ , plus one added point at  $x = 0.3$ ,  $y = 7.6$ , six units above the line
- Left: fit with (red) and without (blue) the added point; right: studentized residuals against fitted values, with the outlier in red

## Problem 5: high-leverage observations, what to check

- **Plot:** studentized residuals  $r^{(i)}$  against leverage  $h^{(i)}$ , with a vertical line at the average  $(p + 1)/n$  and horizontal lines at  $\pm 3$
- **If** points lie far to the right of the average-leverage line, **then** they have high leverage: unusual feature values that can pull the fit towards themselves
- **If** a high-leverage point also has a large residual, **then** it is *influential*: removing it changes the fit a lot
- High leverage alone is not a violation of an assumption; the plot checks whether one point controls the fit
- **Remedy:** check the point, refit without it, and report both fits

## Problem 5: high-leverage observations, example





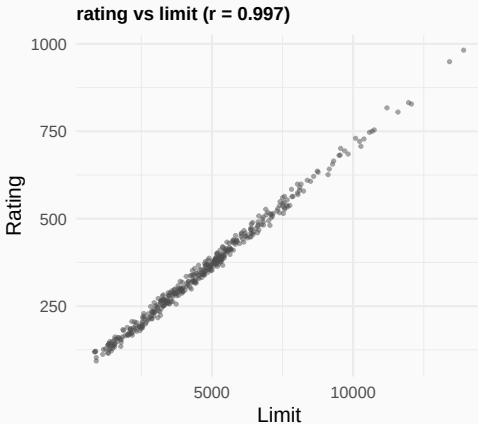
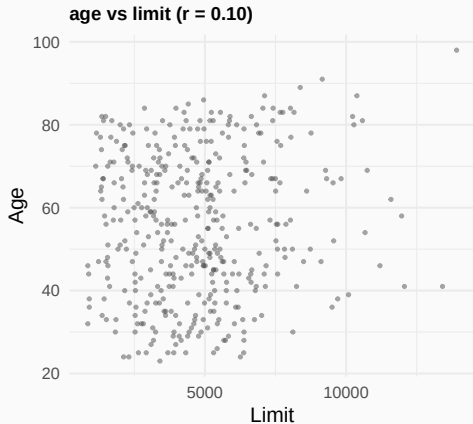
## Problem 5: high-leverage observations, example (cont.)

- Dataset (simulated): the 50 points of Problem 4 without the outlier, plus one added point at  $x = 5, y = 2$ ; the model  $y = \beta_0 + \beta_1 x + \varepsilon$  is fitted to all 51 points
- Left: fit with (red) and without (blue) the added point; right: studentized residuals against leverage
- The added point has the most extreme  $x$ , so its leverage is high; its studentized residual is also large and negative, so it is influential and the fitted line tilts visibly towards it
- Real data, Auto (model of Problem 3): the largest leverage is  $h^{(i)} = 0.095$  (a 225-horsepower car), 9.3 times the average  $4/392$

## Problem 6: collinearity, what to check

- **Plot:** a scatterplot of each pair of features  $x_j$  against  $x_k$ , with the correlation  $r_{jk}$  in the title
- **If** the points lie almost on a straight line ( $|r_{jk}|$  close to 1), **then** the two features carry nearly the same information, and the no-perfect-collinearity assumption is nearly violated
- Confirm with the VIF: values above 5 or 10 signal collinearity
- **Consequence:** unstable coefficients, inflated standard errors, and important features that look insignificant
- **Remedy:** drop one of the redundant features, or combine them

## Problem 6: collinearity, example



## Problem 6: collinearity, example (cont.)

- Dataset: Credit,  $n = 400$  customers; response Balance, with features limit, rating and age
- Left: limit against age,  $r = 0.10$ , a diffuse cloud; right: limit against rating,  $r = 0.997$ , points lying almost on a line
- For Balance on age + limit,  $\text{SE}(\hat{\beta}_{\text{limit}}) = 0.0050$  ( $p < 0.0001$ ); for rating + limit, 0.0638, 13 times larger, with  $p = 0.70$

## Detecting collinearity: variance inflation factor

- Pairwise correlations can miss collinearity among three or more features (multicollinearity); the **variance inflation factor** does not

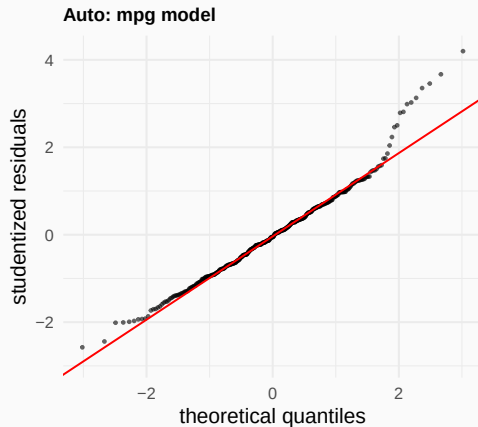
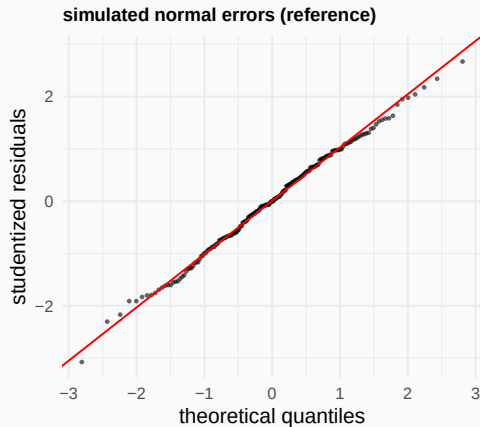
$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1 - R_{x_j|x_{-j}}^2}$$

- $R_{x_j|x_{-j}}^2$  is the  $R^2$  from regressing feature  $x_j$  on all the other features; VIF is at least 1, and values above 5 or 10 signal a problem
- Credit, age + rating + limit: VIF = 1.01 (age), 160.7 (rating), 160.6 (limit); with only age and limit: 1.01 and 1.01
- Remedies: drop one of the redundant features, or combine them (for example, average the standardized features)
- Predictions barely change ( $R^2 = 0.750$  and  $0.746$  for the two models), but individual coefficients and their inference do

## Problem 7: non-normal errors, what to check

- **Plot:** normal QQ plot of the studentized residuals  $r^{(i)}$ : sorted residuals against the quantiles of  $\mathcal{N}(0, 1)$ , with a reference line
- **If** the points follow the line, **then** normal errors are plausible
- **If** both ends bend away from the line, **then** the tails are heavier than normal: there are more extreme residuals than the normal model allows
- **If** only one end bends away, **then** the errors are skewed
- **Consequence:** exact small-sample intervals and tests are unreliable; with large  $n$  the central limit theorem keeps estimates and inference roughly valid
- **Remedies:** transform  $y$ , use robust regression, or bootstrap the inference

## Problem 7: non-normal errors, example



## Problem 7: non-normal errors, example (cont.)

- Left: simulated normal errors,  $n = 200$ ; the points follow the line, which is what a normal QQ plot should look like
- Right: Auto,  $n = 392$ , model of Problem 3; the upper tail rises above the line, with the largest residual at 4.20 against about 3.0 expected under normality; the lower tail is less extreme than normal
- Skewness 0.67 and excess kurtosis 1.56: right-skewed with heavier tails; the largest residuals come from the high-mpg cars of Problem 4



# Regression Performance Measures

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# Measures of the quality of fit

| measure        | definition                                           | comment                                   |
|----------------|------------------------------------------------------|-------------------------------------------|
| MSE            | $RSS/(n - p - 1)$                                    | Squared units of $y$                      |
| RSE            | $\sqrt{MSE}$                                         | Units of $y$ , estimates $\sigma$         |
| MAE            | $\frac{1}{n} \sum_{i=1}^n  y^{(i)} - \hat{y}^{(i)} $ | Units of $y$ , less sensitive to outliers |
| $R^2$          | $1 - RSS/TSS$                                        | Unit-free, never falls with more features |
| Adjusted $R^2$ | $1 - \frac{RSS/(n - p - 1)}{TSS/(n - 1)}$            | Unit-free, penalizes each added feature   |

where  $RSS = \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2$ .

- MSE and RSE punish large errors heavily (squares); MAE weighs all errors linearly, so a few outliers move it less

# Interpreting $R^2$

- $R^2 = 1 - \text{RSS}/\text{TSS} = \text{ESS}/\text{TSS}$  is also the squared correlation between  $y$  and  $\hat{y}$ , and equals  $r^2$  in simple regression
- It lies in  $[0, 1]$  for a model with an intercept, and it never decreases when a feature is added: the larger model can always reproduce the smaller one by setting the new coefficient to zero
- It depends on the spread of the data: the same model fitted where  $x$  (and hence  $y$ ) varies over a wider range has a higher  $R^2$ , and  $R^2$  values of models with different responses (for example mpg and log mpg) are not comparable
- High  $R^2$  does not mean the model is right (check the residuals); low  $R^2$  does not mean it is useless, since real effects can be small relative to noise, for example in survey or behavioural data

# Adjusted R-squared

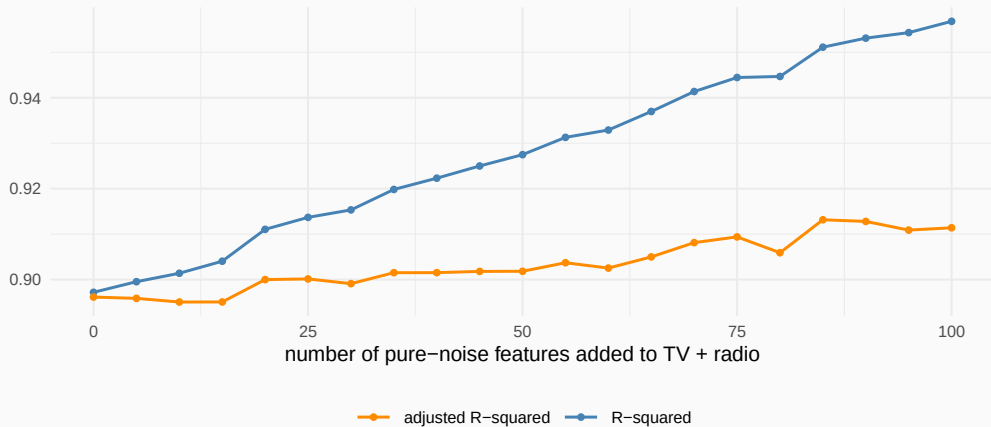
$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)} = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

- It compares two unbiased variance estimates: the residual variance  $\hat{\sigma}^2$  and the variance of  $y_i$ ; the factor  $(n - 1)/(n - p - 1)$  grows with  $p$  and penalizes each added feature
- Adding a feature raises  $R_{\text{adj}}^2$  exactly when the feature's  $|t|$ -statistic exceeds 1 (equivalently, its partial  $F$  exceeds 1); a weak bar, but it stops  $R_{\text{adj}}^2$  from rising for pure noise
- Advertising: adding newspaper has  $|t| = 0.18 < 1$ , so  $R_{\text{adj}}^2$  falls from 0.8962 to 0.8956 while  $R^2$  stays 0.8972

## Adjusted R-squared with useless features

- Dataset: Advertising,  $n = 200$ ; model: sales on TV + radio, with pure-noise features added to it, drawn independently of sales
- Plot:  $R^2$  and  $R_{\text{adj}}^2$  against the number of noise features added cumulatively, in steps of 5 up to 100
- $R^2$  can only rise, here from 0.897 to about 0.95 with 100 noise features;  $R_{\text{adj}}^2$  stays near 0.90 and shows no systematic gain

## Adjusted R-squared with useless features (cont.)



## Comparing regression models on the Advertising data

| model                  | RSE   | RMSE  | MAE   | $R^2$  | adjusted $R^2$ |
|------------------------|-------|-------|-------|--------|----------------|
| TV                     | 3.259 | 3.242 | 2.550 | 0.6119 | 0.6099         |
| TV + radio             | 1.681 | 1.669 | 1.254 | 0.8972 | 0.8962         |
| TV + radio + newspaper | 1.686 | 1.669 | 1.252 | 0.8972 | 0.8956         |
| TV $\times$ radio      | 0.944 | 0.934 | 0.660 | 0.9678 | 0.9673         |

- RMSE and MAE are in thousands of units of sales: the typical error falls from about 3.2 (TV only) to about 0.9 (TV  $\times$  radio)
- Adding newspaper leaves  $R^2$  unchanged (0.8972) but lowers adjusted  $R^2$  (0.8962 to 0.8956) and raises RSE: the penalty exposes a useless feature
- All numbers here are computed on the training data used to fit the models

# Summary

- **Assumptions:** linearity, zero-mean errors, full column rank, uncorrelated errors, constant variance and (for exact inference) normal errors; these may not hold, so they must be checked
- **Residuals:**  $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ , with  $\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$ ; studentized residuals  $r^{(i)} = e^{(i)} / (\hat{\sigma} \sqrt{1 - h^{(i)}})$
- **Diagnostics:** residual plots reveal non-linearity, correlated errors and non-constant variance; studentized residuals flag outliers, leverage flags unusual feature values, VIF flags collinearity
- **Performance:** MSE, RMSE, MAE, RSE,  $R^2$  and adjusted  $R^2$  measure the fit to the training data; RSE and adjusted  $R^2$  account for the degrees of freedom; training error keeps falling as features are added, test error does not



**Questions?**

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